Industrial Power Management Company

UNITY ENERGY

INTELLIGENT OPTIMIZED POWER

EXAMINE ▽ HARVEST △ REFINE ▽ MAP △ EXCHANGE ▽ OPTIMIZE

ENERGY REPORT

2023 DATA

System Value Opportunities

- Precision Measurement
- Intelligent Power Management
- Renewable Energy Exchange

Prepared for;

Foster Farms

Cherry Ave Facility

Fresno, California 93706

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A New Era for Industrial Electricity

The advent of electromagnetic induction in 1831 marked the dawn of a new era in electrical science. Yet, the potential of Reactive Energy, a force inherent in this phenomenon, remained untapped for over a century.

Now, science has allowed technology to advance and unlock the value of Reactive Energy. This breakthrough stems from the understanding that "big" and "small" electricity differ fundamentally, requiring distinct approaches to management.

This groundbreaking system unifies the Energy Field, harmonizing the Utilities' Electrical Energy Supply with the Power Consumed. It intercepts harmonic energy, removes destructive reactive energy, and calms each energy field at the source. This process, powered by "copper computers," establishes Unity Power within the Energy Field, resulting in significant energy savings, enhanced equipment longevity, and a reduced carbon footprint.

This technological advancement has paved the way for a new detailed understanding and lexicon in the field of industrial electricity, challenging long-held assumptions and opening up new possibilities for power management. This will reshape how we think about alternating current electricity and transform the industries that rely on it.

Welcome to the Future of Industrial Electricity

Unity Systems

ELECTRICAL ENERGY BASICS

Foster Farms - CherryAVE Site

UTILITY PROVIDER:
Pacific Gas and Electric Company

SUPPLY PROVIDER:
Shell Energy Solutions

TRANSFOMER LIST Capacity = 13.4MW

SITE CAPACITY 13,400kVA ; 12,342.7A

4- 480v XFMRS	**= 9,000kVA**	**= 10,825.3A**	**= 87.7%**
2- 4160v XFMRS	= 4000kVA	= 555.1A	= 4.5%
2- 240v 200kVA	= 400kVA	= 962.3A	= 7.8%

ENERGY HARVEST OBJECTIVES

MULTI-STAGED TO ACHIEVE THE OPTIMUM ENERGY USE

1 - ENERGY EXAM

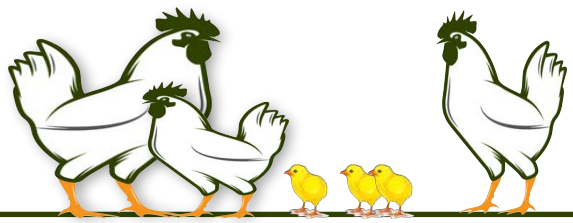
- Install Advanced Metering Systems on all Energy Fields
- Study, Understand, and Report the Customer Site Energy Behavior

2 - POWER MANAGEMENT

- Install Advanced Power Management Devices onto each 480v Energy Field, as needed, to harvest and convert the Reactive Energy into Power.
- Unity currently offers 2 sizes:
  H240 (1000 lbs Copper Computer) and H490 (2000 lbs Copper Computer)

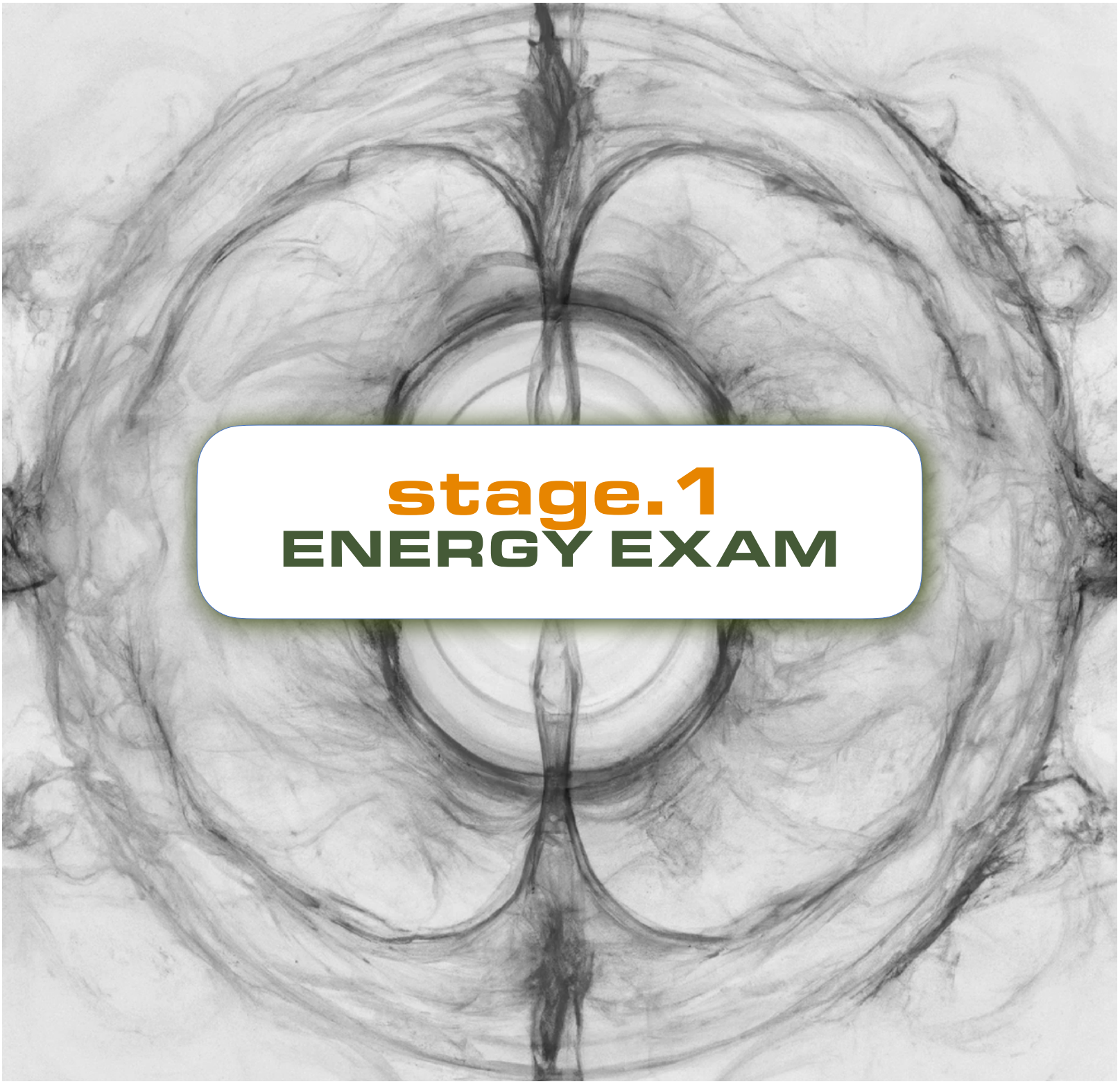
3 - DEMAND EXCHANGE

- Install Unity Solar-LPF Battery System to allow dynamic load-profile following
- Dynamically exchange the Demand and Higher Cost Peak Time Energy with stored Lower Cost Battery energy from lost cost renewable sources
- Monitor, Improve, and Load Balance each Energy Field System using the Latest developments in Artificial Intelligence.



Industrial end-load

ENERGY BEHAVIOR



stage.1
ENERGY EXAM

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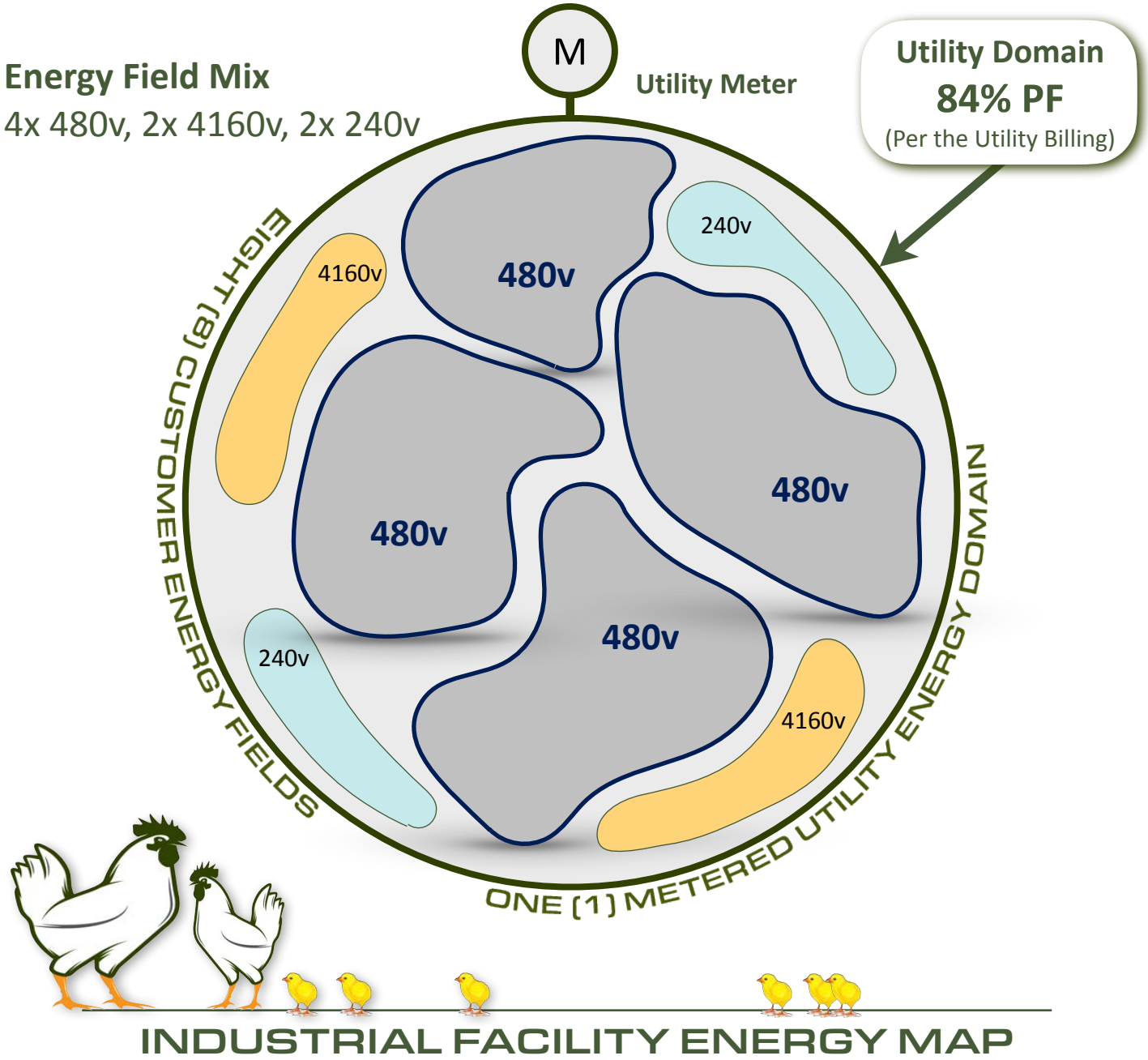
Unity Systems

The Industrial Energy Domain

STEP 1 - Understand and Measure the Facility's Energy Health

- Utilities monitor a single energy domain and meter, despite the presence of eight energy fields within a customer's facility.
- Installation of secure, active streaming metering systems provides detailed insights into the facility's electrical health.
- This approach ensures accurate reporting of energy and CO2e savings to federal and state agencies.



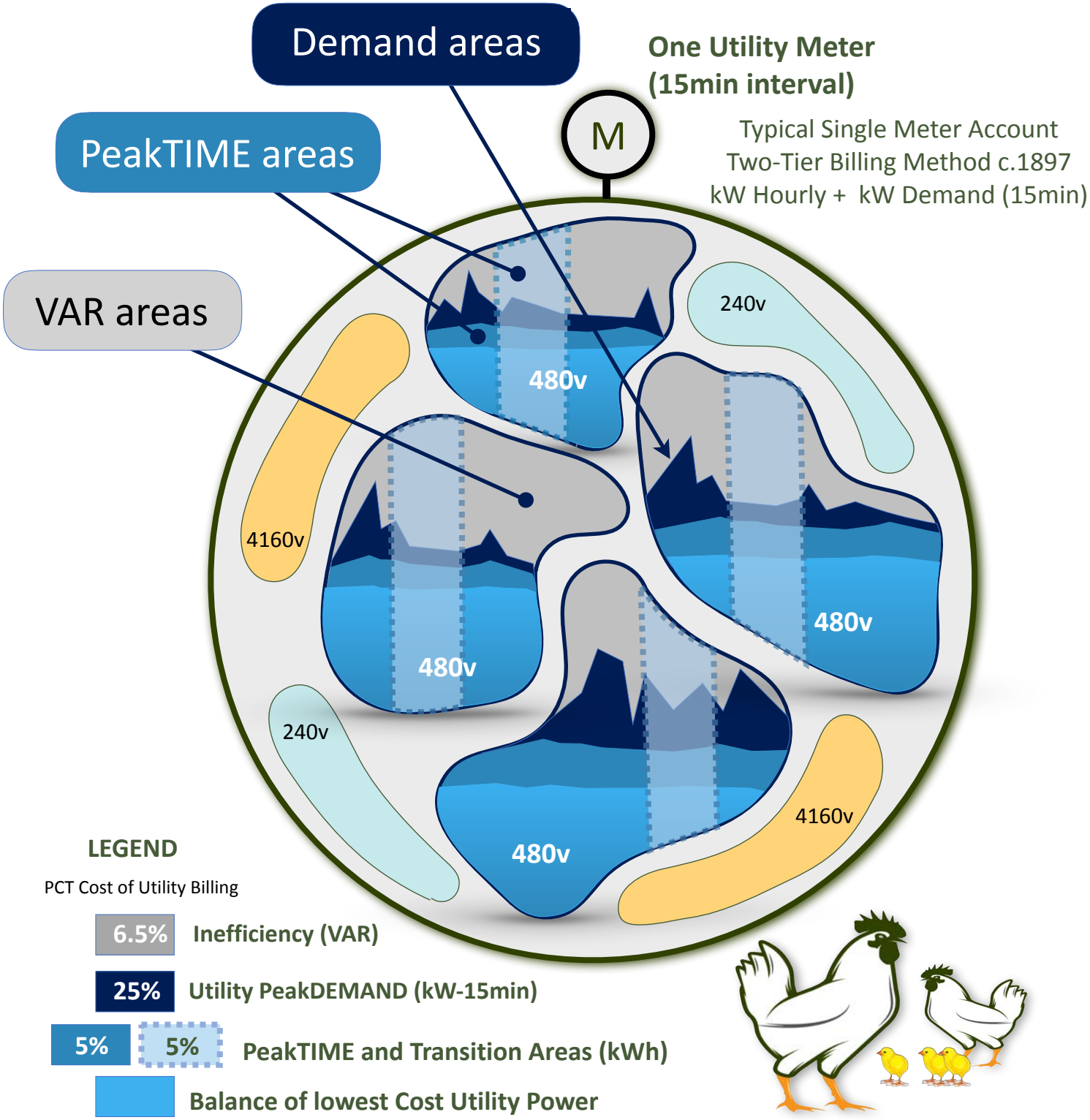
INDUSTRIAL FACILITY ENERGY MAP

Understanding the Energy Domain

Step 2. Three AREAS to Harvest Savings within the 480v Fields

Inefficiencies(VAR), PeakDMD (kW), and PeakTIME(kWh)

These Areas account for over **40% of the Utility Bill Costs.**



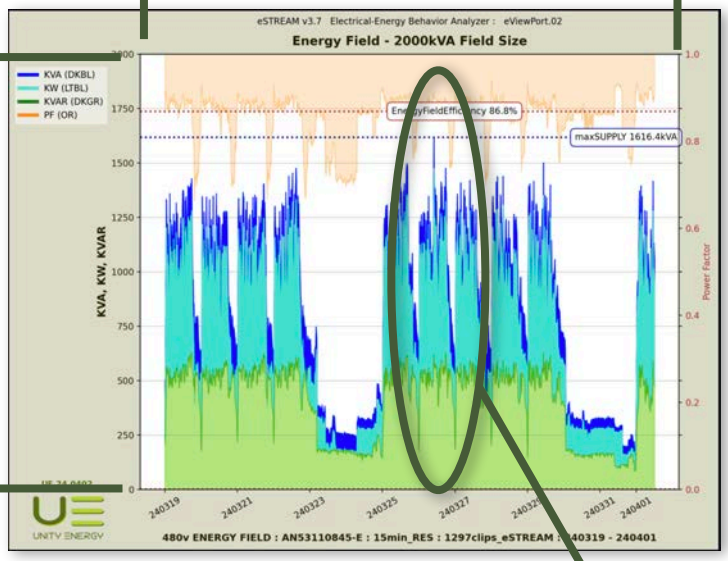
Unity Systems

Understanding the Energy Domain

REAL-WORLD IMAGE OF ONE 480v ENERGY FIELD STREAM

TIMEFRAME
(2 weeks)

FIELD SIZE
(2000 kVA)



UNCONSUMED ENERGY
Destructive Energy created
From the REACTION

REACTIVE ENERGY
Represents the Reactions
From the power transfer

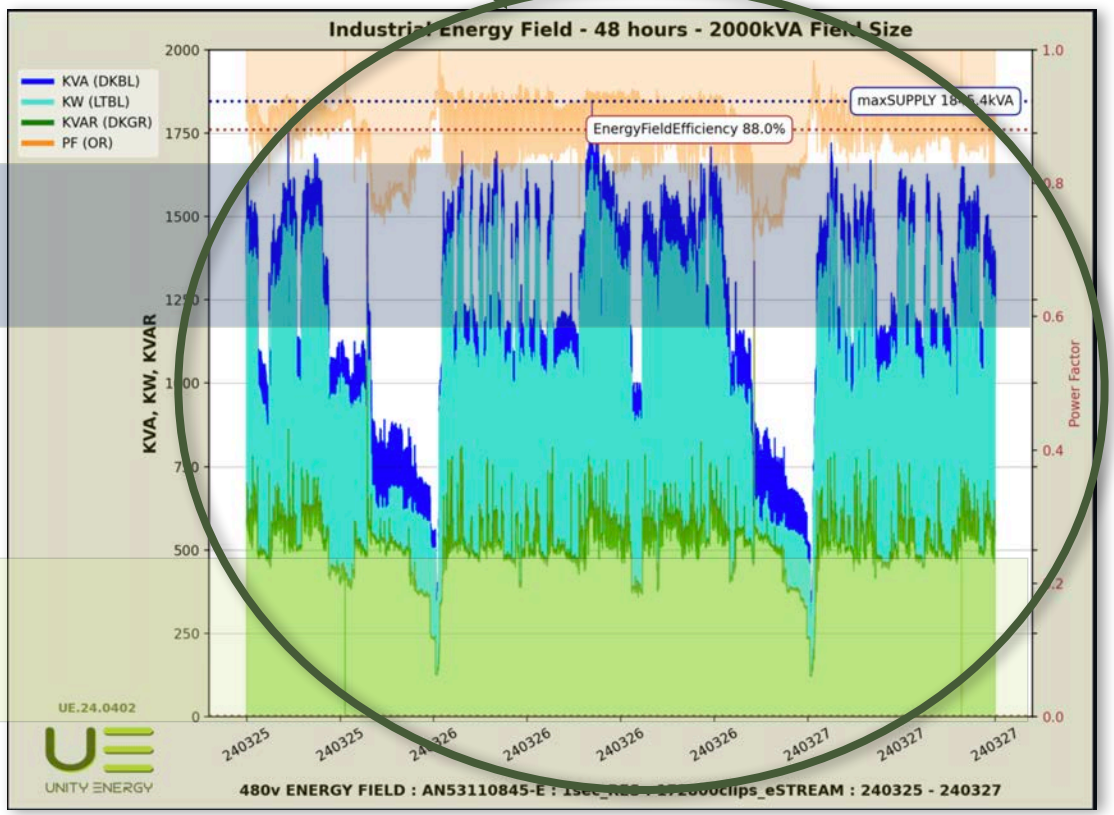
CONSUMED ENERGY

FIELD INEFFICIENCY

The DEMAND ZONE
Accounts for 25%
Of the Utility Billing Cost

kW DEMAND
Behavior Zone

REACTIVE ENERGY



This Reactive Energy
Is caused by poor
energy transfer
Creating Unconsumed Energy (DkBlu) to remain in the Energy Field. This Energy consists of Steady-State Energy (Heat) and Dynamic Energy (Harmonics).

BASELINE Energy

Target Site - basis 2023 Utility Bills

Site Basics

Site Capacity = **13,540A**

Energy Cost per kWh = **$0.2250**

Power Factor = **84.0%**

Three Voltage Energy Fields 240v, 480v, 4160v

480v Energy Occupies 87.7% of the Total Energy Field

Energy Supplied is Equivalent to about 4,120 single-family homes

Energy Performance

Total Energy Cost (Utility) = $700,000 per month

kWh = $525,000 monthly (75%) + kW Demand = $175,000 monthly (25%)

Energy Supplied	= 5,000 kVA per hour
Energy Consumed	= 4,200 kW per hour
Peak DEMAND (15mins)	= 6,900 kW max
Unconsumed Energy (Lost)	**= 800 kVA per hour**
Lost Energy Value (Utility Loss)	**= $1.58M per year**

Energy Lost is Equivalent to about 660 single-family homes

Emissions Performance

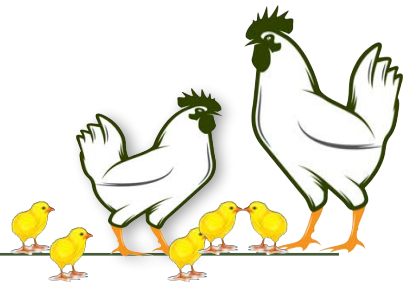
Annual ENERGY - CO2e Emissions = 17,082 MT (metric tons)

Pollution is Equivalent to about 4200 Gas Powered Cars

Bird Performance

Based on 950,000 Birds Processed per week

Energy Cost per bird	= **0.170 cents**
Energy Used per bird	**= 892 Watts**
CO2e Emissions per 1,000,000 birds	**= 348 MT of CO2e**



Industrial end-load

ENERGY BEHAVIOR

stage.2
POWER MANAGEMENT

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Unity Systems

UNITY Energy System Applied

Energy and CO2e Recovery Available

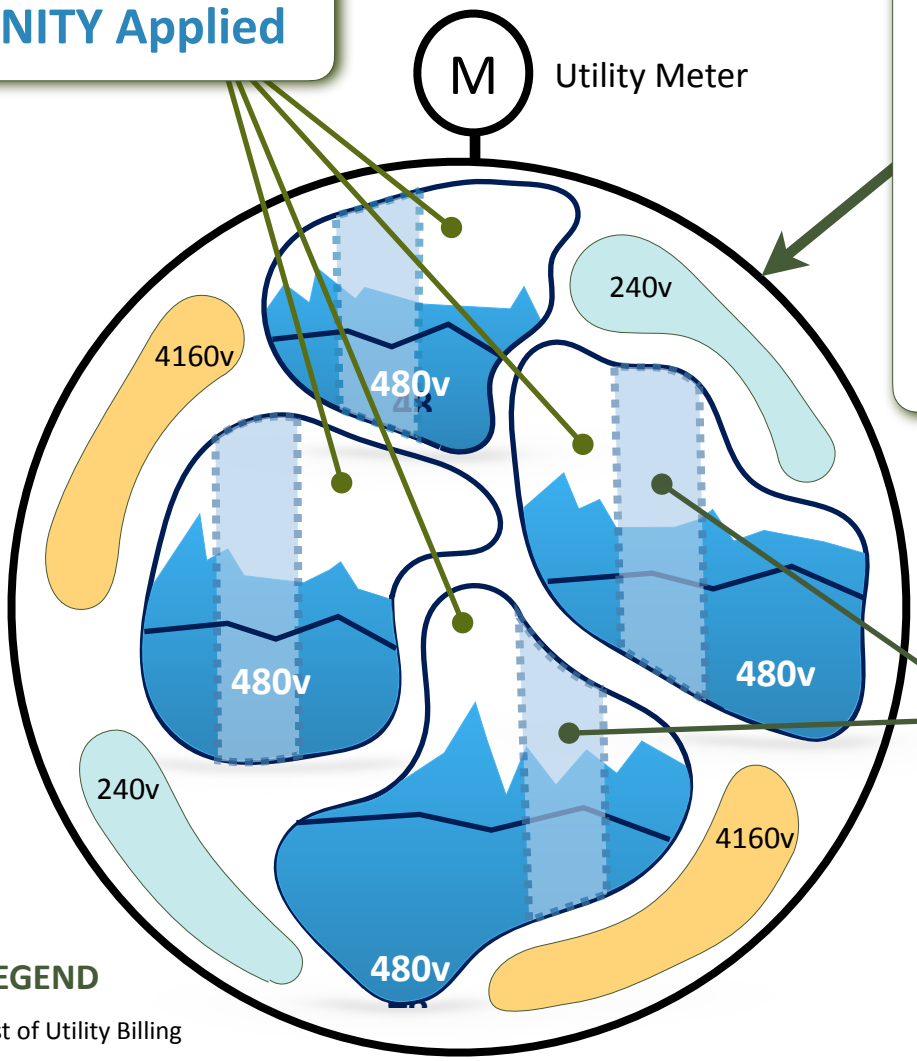
UTILITY Supply Recovery = 750kW/hr (15%)

CO2e Emissions Reduction = 2,650 MTCO2e Annually

480v Energy Fields
99.5% PF
UNITY Applied

Stage 2 - Refine & Optimize
- High-Resolution
- Continuously Recorded
- Continuously Optimized Energy Fields

Stage 3 - Preparation
Clear Definition;
Provides a Precision Load Profile for the Unity Energy Demand and Peak Exchange System.



Peak $$Cost Zones

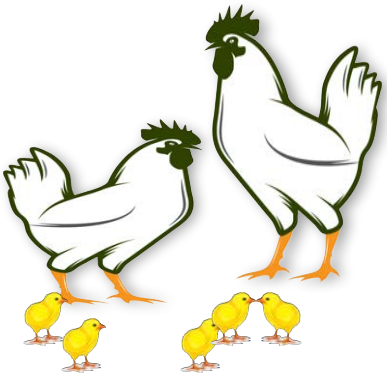
LEGEND

PCT Cost of Utility Billing

5% **PEAK kWh - Cost Zones**

25% **DEMAND and PEAKAreas (kWh) ABOVE Nominal Within the Optimized UNITY POWER**

Nominal below



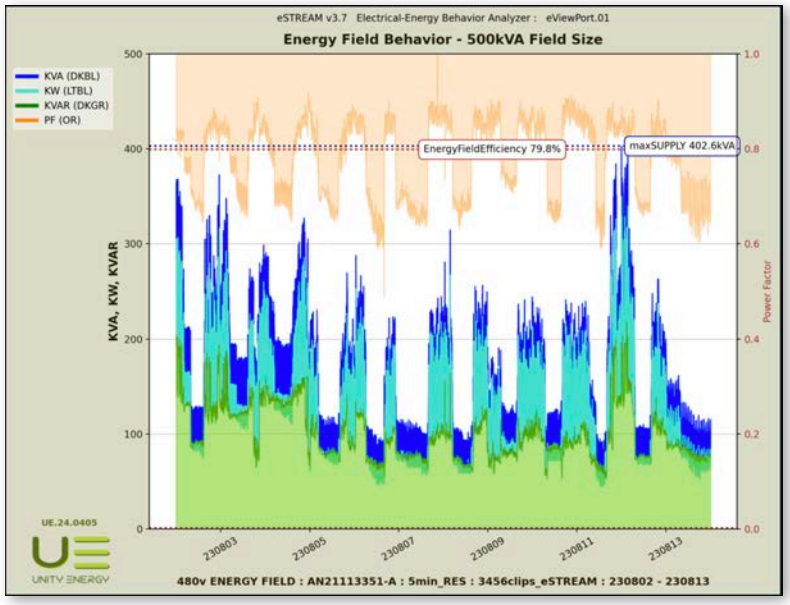
UNITY Energy System Applied

UNDERSTANDING THE APPLICATION

Below is an image showing real-world Energy Fields in large industrial environments. These are the same 8 Energy Fields overlaid at the plant site. The Images Surrounding are Example Installations of both Meter Systems and Unity Power Systems, H490 and H240 w/MPTS Technology.

CAPTURING THE ENERGY CONVERSION - Seeing is Believing

WHEN THE REACTIVE ENERGY (VAR) IS CONVERTED INTO CONSUMABLE END-LOAD POWER, THE EXTRA SUPPLY ENERGY kVA IS NO LONGER REQUIRED.



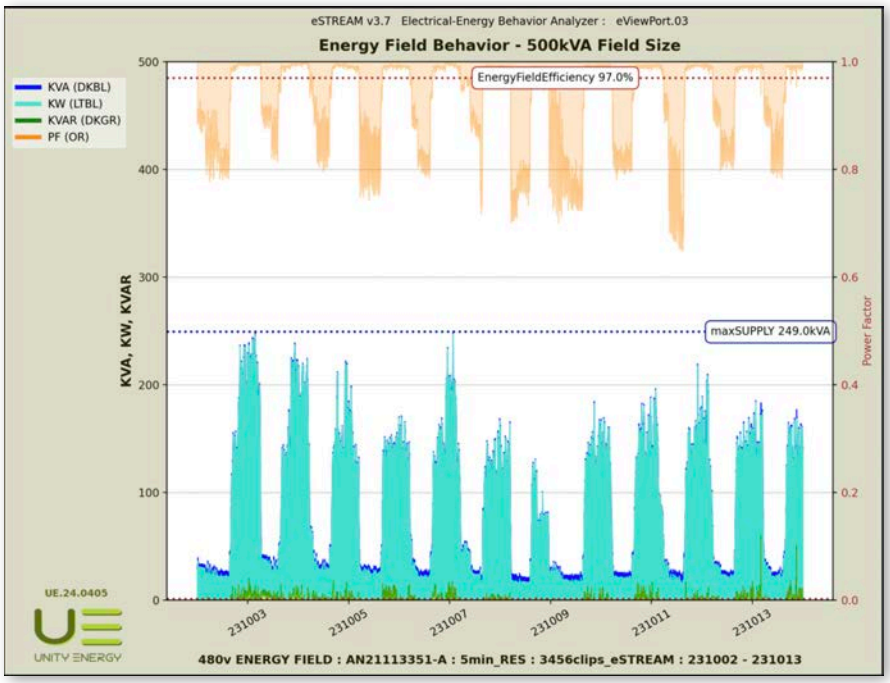
Power Factor Raised 17.7%
SUPPLY Lowered 23.4%

UNCONSUMED ENERGY
Destructive Energy created
From the REACTION

REACTIVE ENERGY
Represents the Reactions
From the power transfer

CONSUMED ENERGY

FIELD INEFFICIENCY



ALMOST ALL REACTIVE ENERGY IS CONVERTED INTO CONSUMABLE POWER. WHICH MEANS THE AMOUNT OF UNCONSUMED WAS REDUCED.

UNITY Systems

POWER MANAGEMENT w/MPTS

VALUE COMPARISONS	*BASELINE*	*UNITY*	*savings*
Energy Cost per kWh	= $0.225	$0.210	**-1.5cents**
Power Factor	= 84.0%	**99.5%**	**+15.5%**
Energy Performance			
kWh Charges /mo	= $525,000	N/C	N/C
kW Demand /mo	= $175,000	$129,500	-$45,500
Utility Energy Cost /mo	= $700,000	**$654,500**	**-$45,500**
Energy Supplied /hr	= 5,000 kVA	**4,250 kVA**	**-750 kVA**
Energy Consumed /hr	= 4,200 kW	N/C	N/C
Peak DEMAND /15mins	= 9,000 kW	**6,450 kW**	**-550kW**

UNITY Energy Recovered = +750 kVA per hour

Recovered Energy Value = $1.38M per year (Utility Gain)

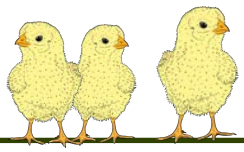
Energy Recovered is Equal to about 615 single-family homes

Emissions Performance	*BASELINE*	*UNITY*	*savings*
CO2e Emission /year	= 17,082 MT	**14,520 MT**	**2,562 MT**

Pollution Removed is Equal to about 630 GasolinePoweredCars

UNITY - Bird Performance			*(LESS PER BIRD)*
Energy	= $.170	**$.157**	**$.013 cents**
Energy Used /1	= 892 W	**830 W**	**62 watts**
CO2e /1,000,000 birds	= 348 MT	**324 MT**	**24 MT**

Based on 950,000 Birds Processed per week



UNITY ENERGY SYSTEM

POWER MANAGEMENT

With TransPower MPTS Technology

BEFORE
UNITY
SYSTEMS

AFTER
UNITY
SYSTEMS

FACILITY
Energy Opportunity

BASED ON 2023 DATA

Base Energy Cost	Base CO2e Debt
$8.4M /yr	**17,000 MTCO2e /yr**

Opportunity Value (Reductions)

20.8 % Achievable Benefits		14.7% Lower Carbon Footprint
$1.75M /yr	:	**2500 MTCO2e /yr**

Measurable Target Savings Annually

$500k	:	**$1.25M**	:	**2,500 MTCO2e**
CUSTOMER		UTILITY OFFSET		EMISSIONS
Lowers Billing		Reduces Fossil Fuel		Reduces CarbonFootprint

Payback of less than 4.0 years

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UNITY Summary

INVESTMENT and RETURN

Achieving a Unity Energy Environment

UNITY SYSTEM COST DETAILS

$1.8M - Unity Site System, plus Installation

Maintenance, Service, Energy Monitoring, and Reporting is separate depending on the financing program

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UNITY SYSTEM VALUE

6.5% - Utility Billing Savings

$ 545,000 - Estimated Savings per Year ($13.6M/25yr Life)

3.3 years - Investment Payback

OTHER Benefits

Reactive Energy Conversion

6.57GW/yr - Utility Supply Reduction-kVA (0.750MW/hour)

$ 1.38M - Energy Displacement Avg. $0.21c /per kWh

Reduced CO2e Emissions

2,562 MT/yr - Carbon Footprint Reduction (64,050 MT/25yr)

$75,000/yr - California Carbon Credit ($20-40 CA2024)

Removal of End-Load Harmonics
Increased Reliability and Extended Component Life
Reduced Electrical Maintenance and Operational Costs.

Industrial end-load

ENERGY BEHAVIOR

stage.3
ENERGY EXCHANGE

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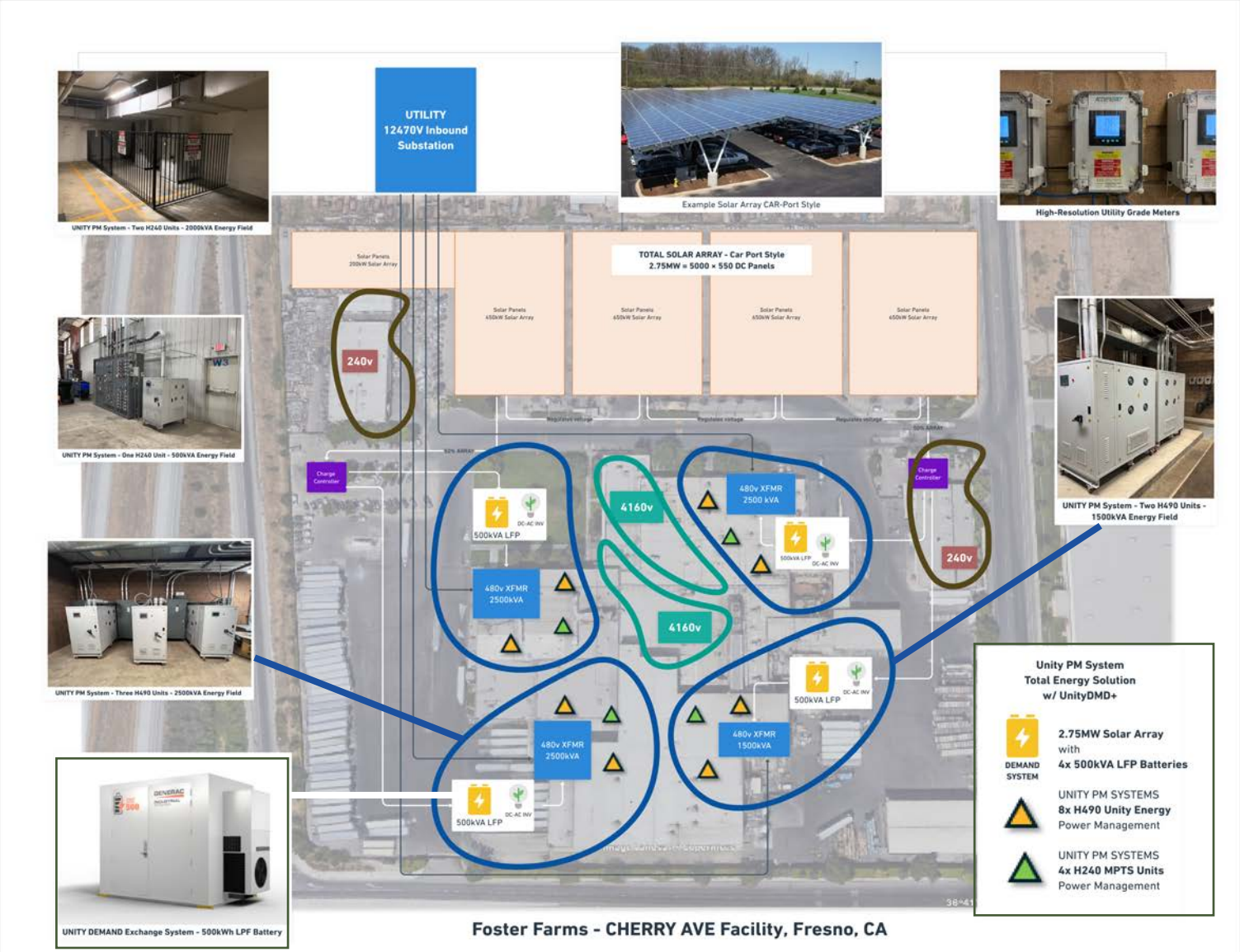
Intelligent Solar-Battery System

Harvesting the Energy Gold (California DEMAND energy)

Once each Energy Field is metered (stage1) and optimized (stage2); we can begin mapping a new clean energy field, and extracting the load profile. This new load profile is used together with Unity's proprietary software to dynamically map and algorithmically control and share the intelligent data with the LFP Battery controllers. —The Solar and Batteries are inside the Utility Bubble.

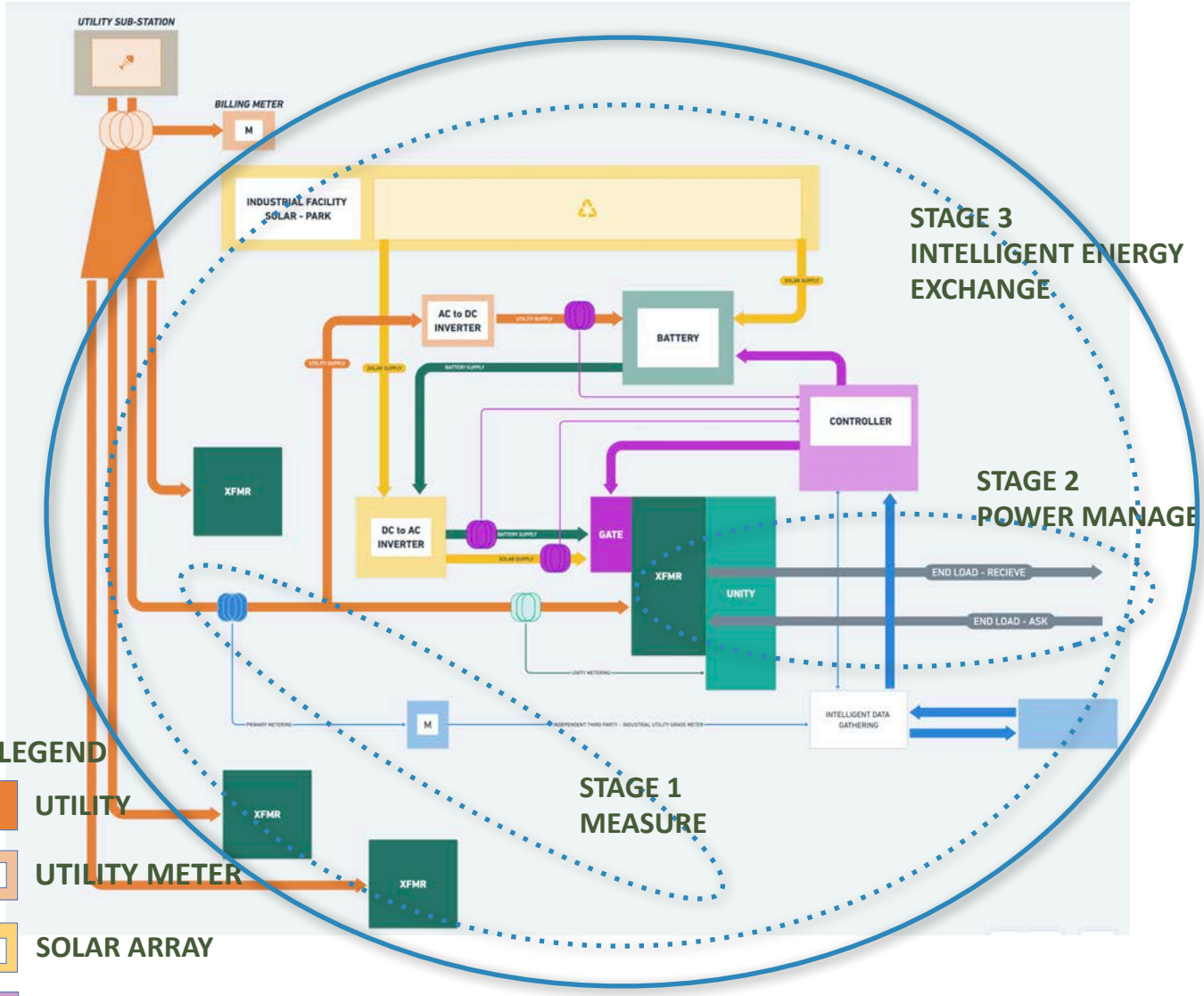


Unity Demand Systems

Intelligent Solar-Battery System

SOLVING OUR INDUSTRIAL DEMAND COST

Below is an image showing schematically how the UNITY Demand System would be applied to each 480v Industrial Transformer. Building on the advanced stages of Measurement, In-Situ Refinement and Overall System Energy Optimization



ENERGY LEGEND

- UTILITY
- M — UTILITY METER
- SOLAR ARRAY
- C — CONTROLLER/INVERTER
- BAT — 500kWh LFP BATTERY
- M — HIGH-RES STREAMING METER
- XFMR — TRANSFORMER WITH UNITY PM
- STANDARD TRANSFORMER

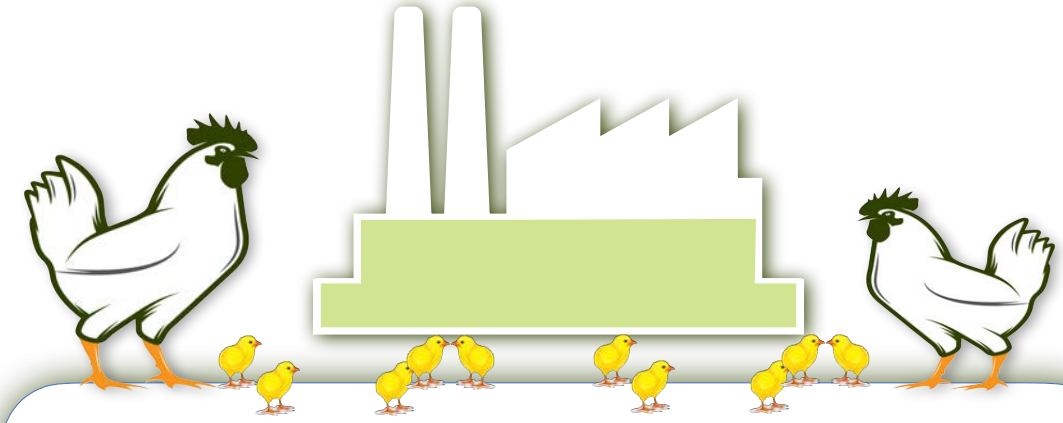
UTILITY Energy Domain

EVERYTHING IS INSIDE THE SINGLE METER UTILITY DOMAIN

UNITY ENERGY SYSTEM

ENERGY EXCHANGE

With TransPower MPTS Technology



FACILITY

Energy Opportunity

BASED ON 2023 DATA

Base Energy Cost	Base CO2e Debt
$8.4M /yr	**17,000 MTCO2e /yr**

Opportunity Value (Reductions)

Almost 50% Achievable Benefits — 26.5% Lower Carbon Footprint

$4.0M /yr : 4,500 MTCO2e /yr

Measurable Target Savings Annually

$2.00M	:	**$1.25M**	:	**4,500 MTCO2e**
CUSTOMER		UTILITY OFFSET		EMISSIONS
Lowers Billing		Reduces Fossil Fuel		Reduces CarbonFootprint

Payback of less than 10 years

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CONTACT DETAILS

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After almost 200 years, science has finally allowed humans to quietly develop a proven technology class of Products that have become the cornerstone of Unity Systems.

Unity Energy offers the first complete Industrial Power Management System that continuously optimizes each 480v Energy field, approaching near-perfect UNITY without using any auxiliary conditioning power.

Our system dynamically converts the unconsumed reactive energy to consumable power at the end loads. This behavior can reduce the amount of Utility Energy required to create the Facility Power by almost 30%.

Our system also raises the intellectual awareness of this critical commodity, enabling companies to become more accountable and better stewards of the Energy we consume for our Electricity needs.

Welcome to the Future of Industrial Electricity

UNITY ENERGY with patented MPTS Technology

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